



UNIVERSITÀ
DI TRENTO

Software Engineering

Course Introduction

Lecture1 - 24/02/2026

Prof. Sandro Fiore

Department of Information Engineering and Computer Science

University of Trento

a.a. 2025-2026

About me and the HPCI Lab @ UNITN



UNITRENTO
DIGITALUNIVERSITY



Sandro Luigi Fiore

Member

Open Science Committee

Associate professor

Department of Information Engineering and Computer Science

Courses:

HPC for Data Science

Software Engineering

Data Science

Business Analytics

GreenOps

Expertise: Artificial Intelligence Big data Climate change Climate informatics Cloud computing Computation

Data analytics Data management systems Data mining Data science Distributed systems

High performance computing Middleware Open science Scientific computing

High Performance Climate Informatics Laboratory

R&D on Computational & Data Sciences applied to **Climate Change** and focusing on:

Scientific Data Management

HPC, Big Data and AI convergence

Workflow & Provenance Management

Computational Sustainability

Course objective

The objective of the course is about providing the students concepts, approaches and techniques for the analysis, the design, and the development of the software



EVENTI

TRENTINO ALTO ADIGE

Trentino Alto Adige Eve...

Trento, “100 progetti software per la Città di Trento”: premiati gli studenti per le migliori app per la città

Alla premiazione “100 progetti software per la Città di Trento,” giovani ingegneri informatici hanno presentato soluzioni per segnalare disservizi urbani e monitorare impianti fotovoltaici, contribuendo alla smart city.

Progetto d'esame: tema

«Ideare la Trento del futuro»



COMUNE
DI TRENTO



UNIVERSITÀ
DI TRENTO

Progetto d'esame: creatività

Ampio spazio alla creatività

- Aspetti cross-cutting ('efficiente', 'smart', 'innovativa', 'sostenibile' etc.)
- Il corso prevede già da questa settimana concetti di **Design Thinking**



Progetto d'esame: meetings

Momenti di incontro:

- (i) oggi
- (ii) presentazione delle idee progettuali
- (iii) entro la fine dell'anno accademico con una premiazione finale sui due Corsi di Laurea (ICE,INF)



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Oltre il progetto: opportunità di tirocini/tesi

Sarà previsto inoltre un seminario a fine corso, in cui il Comune di Trento presenterà opportunità per percorsi di internship e tesi su tematiche specifiche





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Lecturers

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- Prof. Chiara di Francescomarino
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Lectures (I)

- **In presence**
 - Tuesday 10.30-13.30 (A210)
 - Wednesday 08.30-11.30 (A208)
 - Thursday 8.30-10.30 (A222)

	lunedì	martedì	mercoledì	giovedì	venerdì
08.30-09.30			Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A208	Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A222	
09.30-10.30			Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A208	Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A222	
10.30-11.30		Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A210	Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A208		
11.30-12.30		Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A210			
12.30-13.30		Ingegneria del Software <i>Sandro Luigi Fiore / Chiara Di Francescomarino</i> Aula A210			

Lectures (II)

Contents:

- Theoretical
- Practical
- Project activity

You will learn about several **tools** and deliver a **project** by the end of the course with intermediate milestones

Lecture (III)

Effective learning and full achievement of the expected results are facilitated by **basic knowledge of programming** and the development of object-oriented software (*we'll provide basics of **JavaScript** and we'll guide you to the use of the different **tools***)

Further knowledge that can facilitate the understanding of course content is about **database** and **data management** (*we'll provide basics of **MongoDB***).

Course Material

- **Slides** and **additional support material** will be provided for exams and made available on Moodle
- Additional **reading assignments** (see next slides) will be provided too
- **Textbook**: Ingegneria del software
Ian Sommerville. Pearson Printice Hall

Reading (and thinking) assignments

- Are given almost every week
- Most are background and informational
- Some are relevant to the current week's class (i.e. follow up reading)
- Others are relevant to following week's class (i.e. pre-reading)
- We will discuss these in class to stimulate discussions/questions

Exam (Project-based)

- **Project activity throughout the course**
 - Students will work in teams
 - Team members: 3 students
 - Registration form: will be provided soon
 - We'll help you finding teammates (if needed)
 - **Several deliverables are planned**, to be submitted via online form by the due deadline. They include:
 - Documentation, Code, Self-evaluation report

Project calendar

- Deliverable D1 → 27/03/2026
- Pitch della vostra idea → 01/04/2026
- Deliverable D2 → 24/04/2026
- Deliverable D3 → 15/05/2026 (Sprint1 – 3 settimane)
- Deliverable D4 → 07/06/2026 (Sprint2 – 3 settimane)

Project questions

- **Questions**

- To submit a question about the project activity, please use the form here:

<https://docs.google.com/forms/d/e/1FAIpQLSd7lqqwPTfIQ6X8QpvsKXKY1NPM5nmqExA6FV-ve5MHBMAc4w/viewform>

- FAQs will be published on a gdoc

Project: Programming Language & Tools

- Programming languages
 - Javascript
- Tools (just some examples)
 - Editor/IDE
 - Git and github
 - ...
 - (more during the hands-on lectures)

Feedback form

- There will be a ***feedback form*** open throughout the entire course
 - Please provide any comment about:
 - Pros
 - Cons
 - Aspects that were not clear enough during the class
 - Any other feedback you think can be relevant for the course
- Feedback will be collected anonymously
- The feedback form is available here:
<https://docs.google.com/forms/d/e/1FAIpQLScOmS0cpVoXbpABTkVHscLb1z--11ScOh1MSyUnk9u38MUWsA/viewform>

Questions so far?

Course objective

The objective of the course is about providing the students concepts, approaches and techniques for the analysis, the design, and the development of the software

Learning objectives (I)

At the end of the course, the student will be able to:

1. define the most suitable **development process** and **model** for the analysis and the design of the software under consideration;
2. represent with the different types of **diagrams** the **requirements**, the **architecture**, the **interaction** among the components, and the **deployment** of the system;
3. define and manage in a consistent and complete way the **functional** and **non functional** requirements of a software system highlighting priorities and criticalities.
4. define the **architecture of a software system** adopting architectural styles and **principles** of **refinement**, **abstraction**, **modularity**, **functional independency** and **refactoring**;

Learning objectives (II)

At the end of the course, the student will be able to:

5. **design a component of a software system** identifying the control type and following the step-by-step process for the respect of the general design principles;
6. **describe requirements** and implement them in an **agile manner**, with **short development cycles** to ensure regular and **frequent delivery** of software to users;
7. define and document a **web service compliant with RESTful directives**, collaborate into the development of the source code, **testing** the code, and **automating** the **deployment** of the software with a **continuous integration process**.

Course Topics

- Course introduction
- Design thinking
- Software development process
- Modelling languages
- Agile Software Development
- API & API economy
- Project activity (design and implementation milestones + code)

Software development process

- *A set of activities, rules, guidelines, that drive and dictate how a team goes about designing, developing, delivering, and maintaining software.*
- For example, it prescribes **how to build teams**, how to **time releases**, **when to perform which activity**, **how often** the **team should meet**, etc.

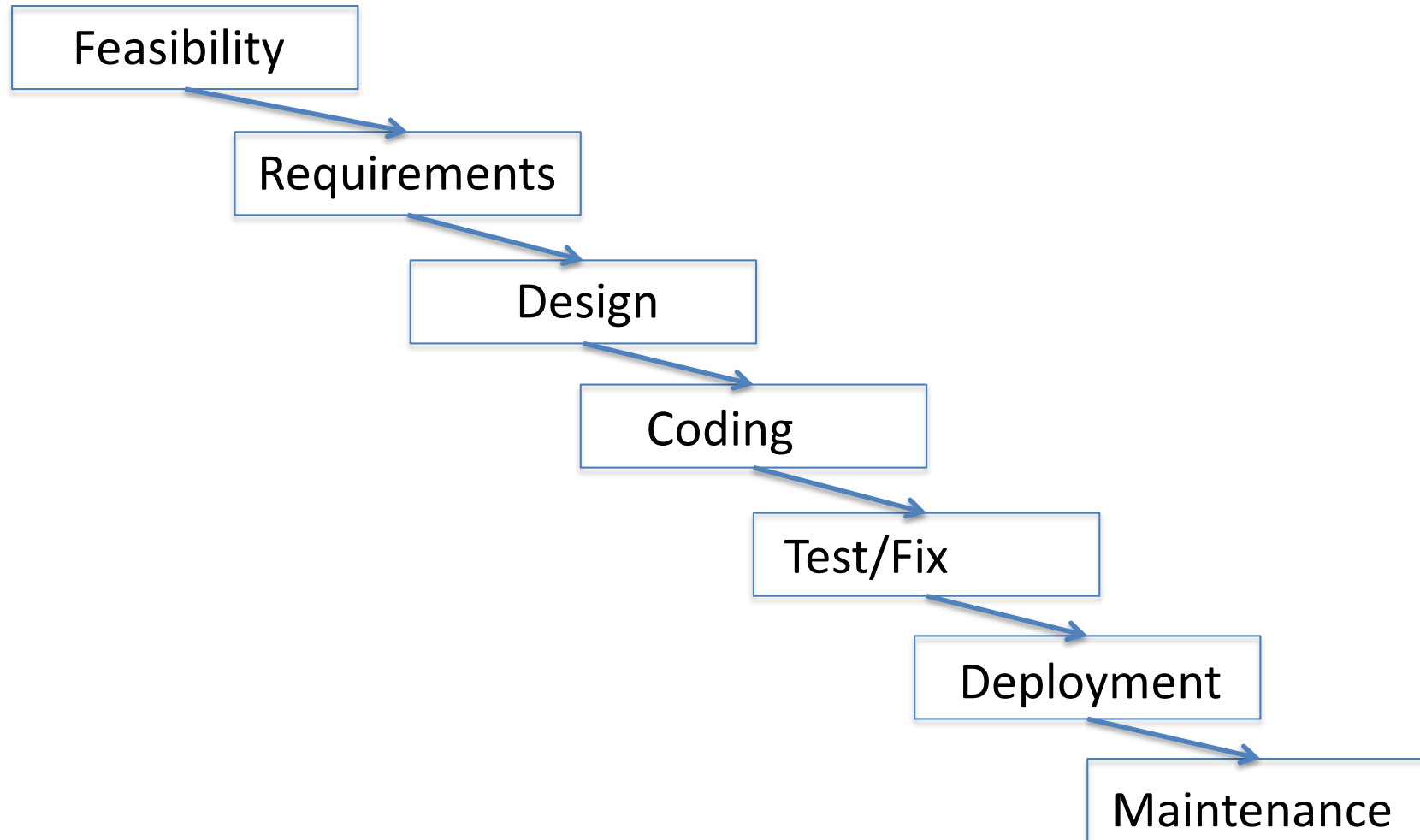
Why do we need it?

- Because we KNOW, by experience, that without a process we WILL:
 - Deliver software that does things wrong
 - Deliver software that does the wrong things

but also:

- Deliver software that is expensive to maintain
- Deliver late
- Deliver at higher operating costs

Waterfall model



Going agile

*Instead of **large group** spending a **long time** building a **big thing**, we have a **small team** spending a **short time** building a **small thing** but **integrating regularly** to see the **whole**.*

– KANBAN VS SCRUM, by Henrik Kniberg

Agile Manifesto

Manifesto for Agile Software Development

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

Individuals and interactions *over* processes and tools
Working software *over* comprehensive documentation
Customer collaboration *over* contract negotiation
Responding to change *over* following a plan

That is, while there is value in the items on the right, we value the items on the left more.

AUTHORS

Kent Beck
Mike Beedle
Arie van Bennekum
Alistair Cockburn

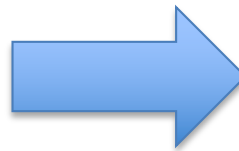
Ward Cunningham
Martin Fowler
James Grenning
Jim Highsmith

Andrew Hunt
Ron Jeffries
Jon Kern
Brian Marick

Robert C. Martin
Steve Mellor
Ken Schwaber
Jeff Sutherland
Dave Thomas

Principle #1

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.



Principles behind the Agile Manifesto

We follow these 12 principles:

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage.

Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale.

Business people and developers must work together daily throughout the project.

Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done.

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

Working software is the primary measure of progress.

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

Continuous attention to technical excellence and good design enhances agility.

Simplicity--the art of maximizing the amount of work not done--is essential.

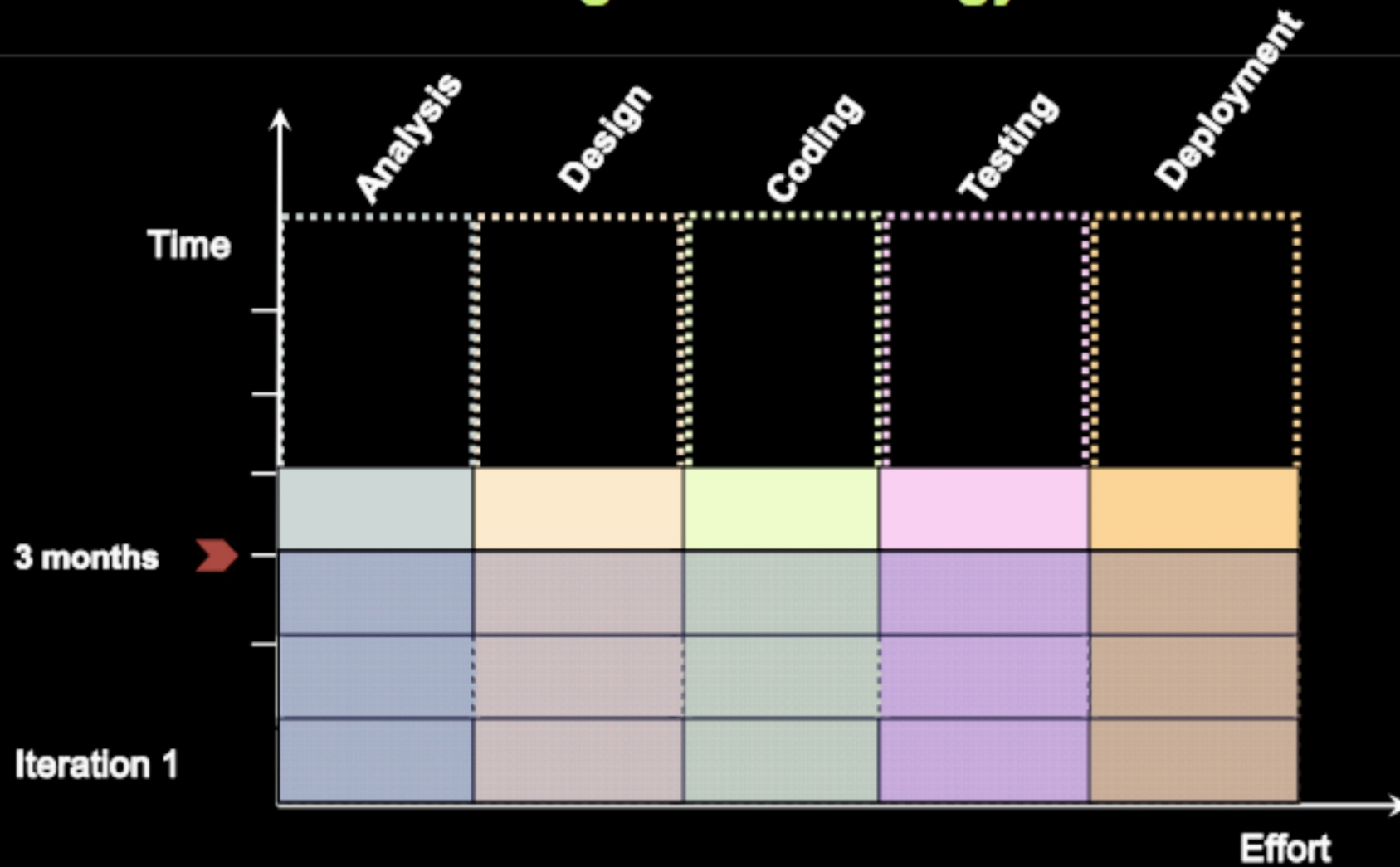
The best architectures, requirements, and designs emerge from self-organizing teams.

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Waterfall Methodology

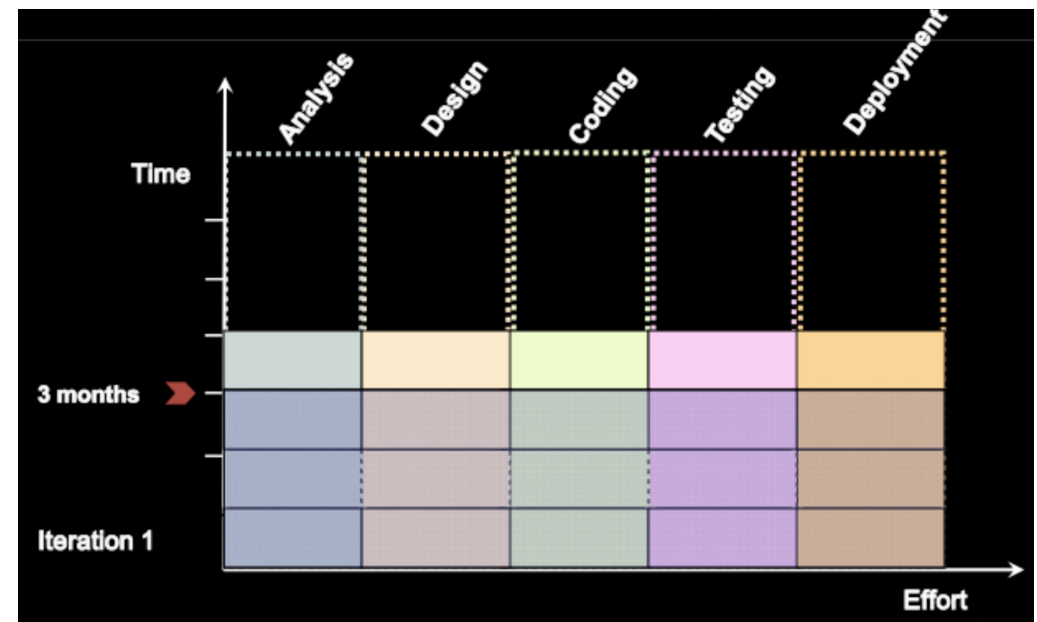
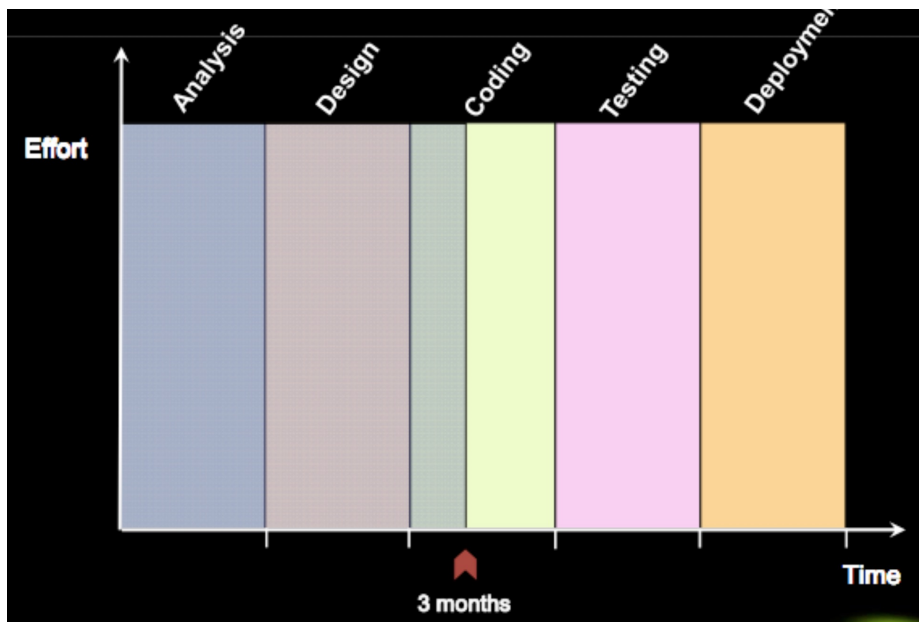


Agile Methodology



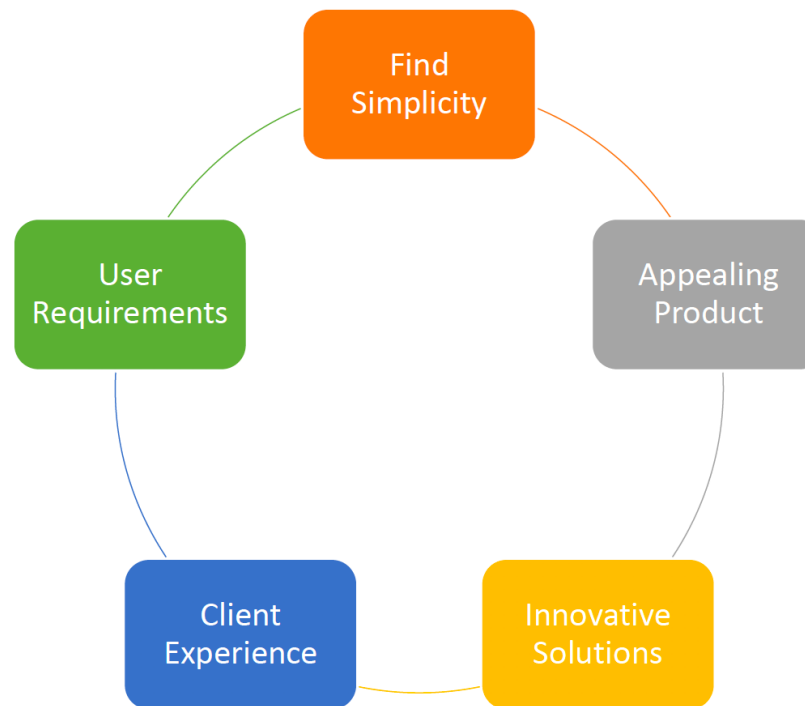
Waterfall vs agile methodologies

What kinds of inherent consequences the two approaches lead to?

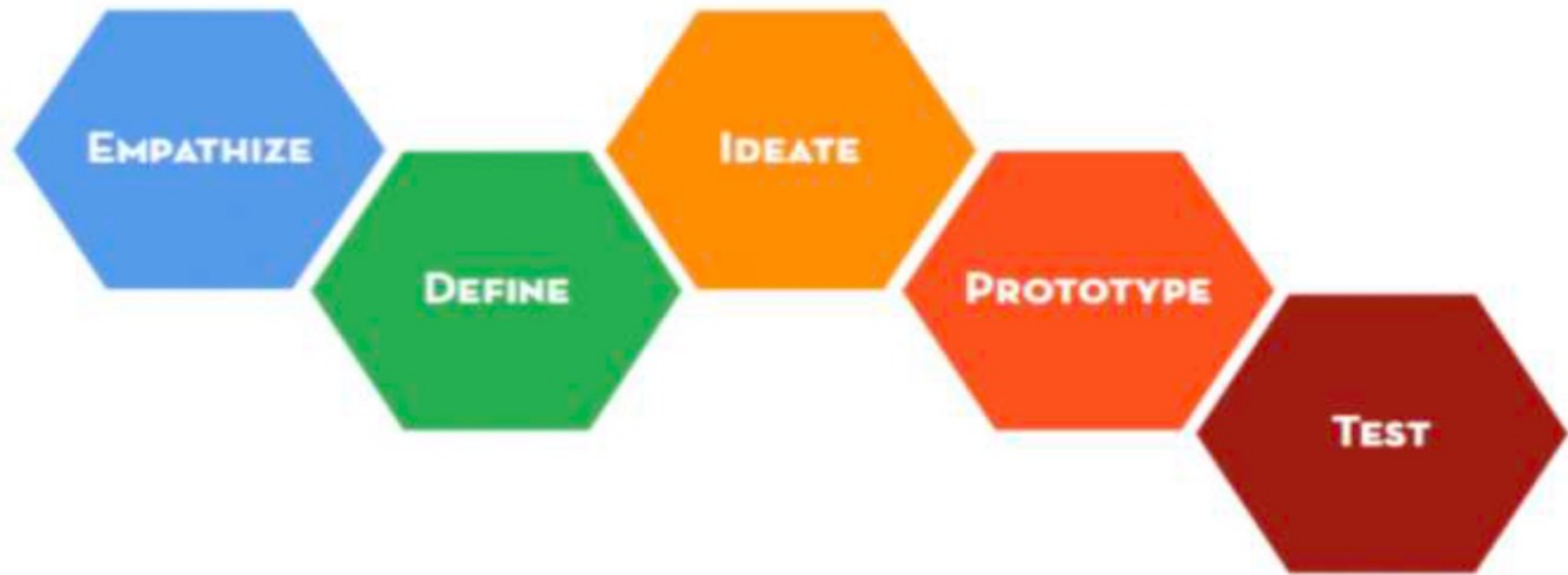


Design Thinking

*Design thinking is a methodology for finding **simplicity in complexity**, improving **quality of experience** with the **designed products** and **serving the needs of customers** by addressing the **target problem** faced by them*



The five-step process



Application Programming Interface

- *An Application Programming Interface (API) defines how other programs can interact with your software.*
- The API provide a specification
- It does not relate to the actual implementation (information hiding)
- **Stability** is a key element for an API

Application Programming Interface

Exercise:

Have you ever used some specific API in your desktop or web applications?

Let's make a list...

API examples: Library

Chapter 6 C API Function Overview

The following list summarizes the functions available in the C API. For greater detail, see the descriptions in [Chapter 7, C API Function Descriptions](#).

- mysql_affected_rows(): Returns the number of rows changed/deleted/inserted by the last UPDATE, DELETE, or INSERT query.
- mysql_autocommit(): Toggles autocommit mode on/off.
- mysql_change_user(): Changes the user and database on an open connection.
- mysql_character_set_name(): Returns the default character set name for the current connection.
- mysql_client_find_plugin(): Returns a pointer to a plugin.
- mysql_client_register_plugin(): Registers a plugin.
- mysql_close(): Closes a server connection.
- mysql_commit(): Commits the transaction.
- mysql_connect(): Connects to a MySQL server. This function is deprecated; use mysql_real_connect() instead.

<https://dev.mysql.com/doc/c-api/8.0/en/c-api-function-overview.html>

API examples: Web API

Overview

The Instagram Basic Display API is an HTTP-based API that apps can use to get an Instagram user's profile, images, videos, and albums.

Components

Base URLs

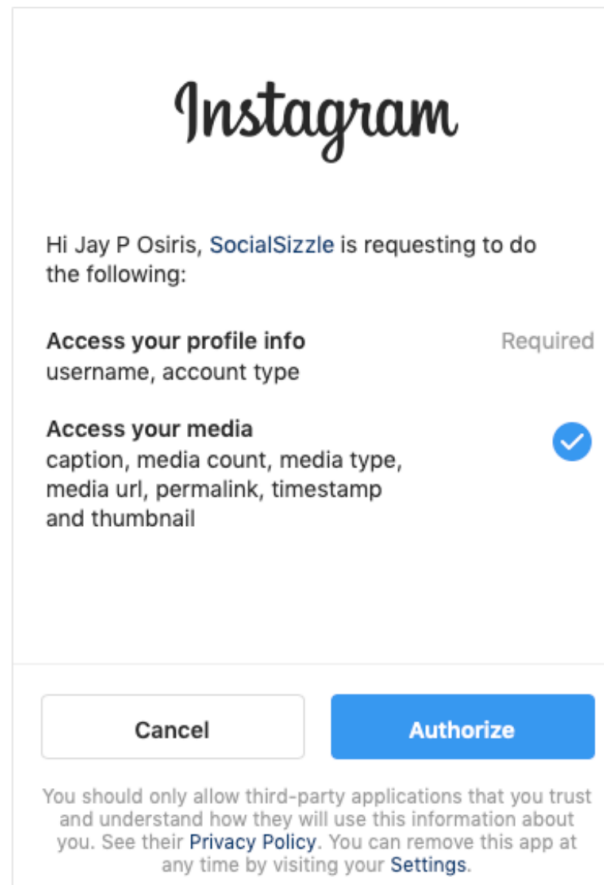
- api.instagram.com — for getting [Instagram User Access Tokens](#)
- graph.instagram.com — for getting Instagram user profiles and media

<https://developers.facebook.com/docs/instagram-basic-display-api/overview>

Instagram example

Authorization Window

The Authorization Window allows your app to get [Authorization Codes](#) and [permissions](#) from app users. Authorization Codes can be exchanged for [Instagram User Access Tokens](#), which must be included when querying an app user's profile or their media.



The image shows a screenshot of the Instagram authorization window. At the top is the Instagram logo. Below it, a message says "Hi Jay P Osiris, SocialSizzle is requesting to do the following:". There are two permission items: "Access your profile info" (username, account type) marked as "Required", and "Access your media" (caption, media count, media type, media url, permalink, timestamp and thumbnail) which has a blue checkmark icon. At the bottom are "Cancel" and "Authorize" buttons. Below the buttons is a disclaimer: "You should only allow third-party applications that you trust and understand how they will use this information about you. See their Privacy Policy. You can remove this app at any time by visiting your Settings."



To implement the Authorization Window, refer to our [Getting Access Tokens](#) guide.

Get Access Tokens and Permissions

This guide explains how to use the [Authorization Window](#) to get [short-lived Instagram User Access Tokens](#) and [permissions](#) from Instagram users.



Step 1: Get Authorization

The [Authorization Window](#) allows app users to grant your app [permissions](#) and [short-lived Instagram User Access Tokens](#). After a user logs in and chooses which data to allow your app to access, we will redirect the user to your app and include an [Authorization Code](#), which you can then exchange for a short-lived access token.

To begin the process, get the Authorization Window and present it to the user:

```
https://api.instagram.com/oauth/authorize
?client_id={instagram-app-id}
&redirect_uri={redirect-uri}
&scope={scope}
&response_type=code
&state={state}           //Optional
```

Query String Parameters

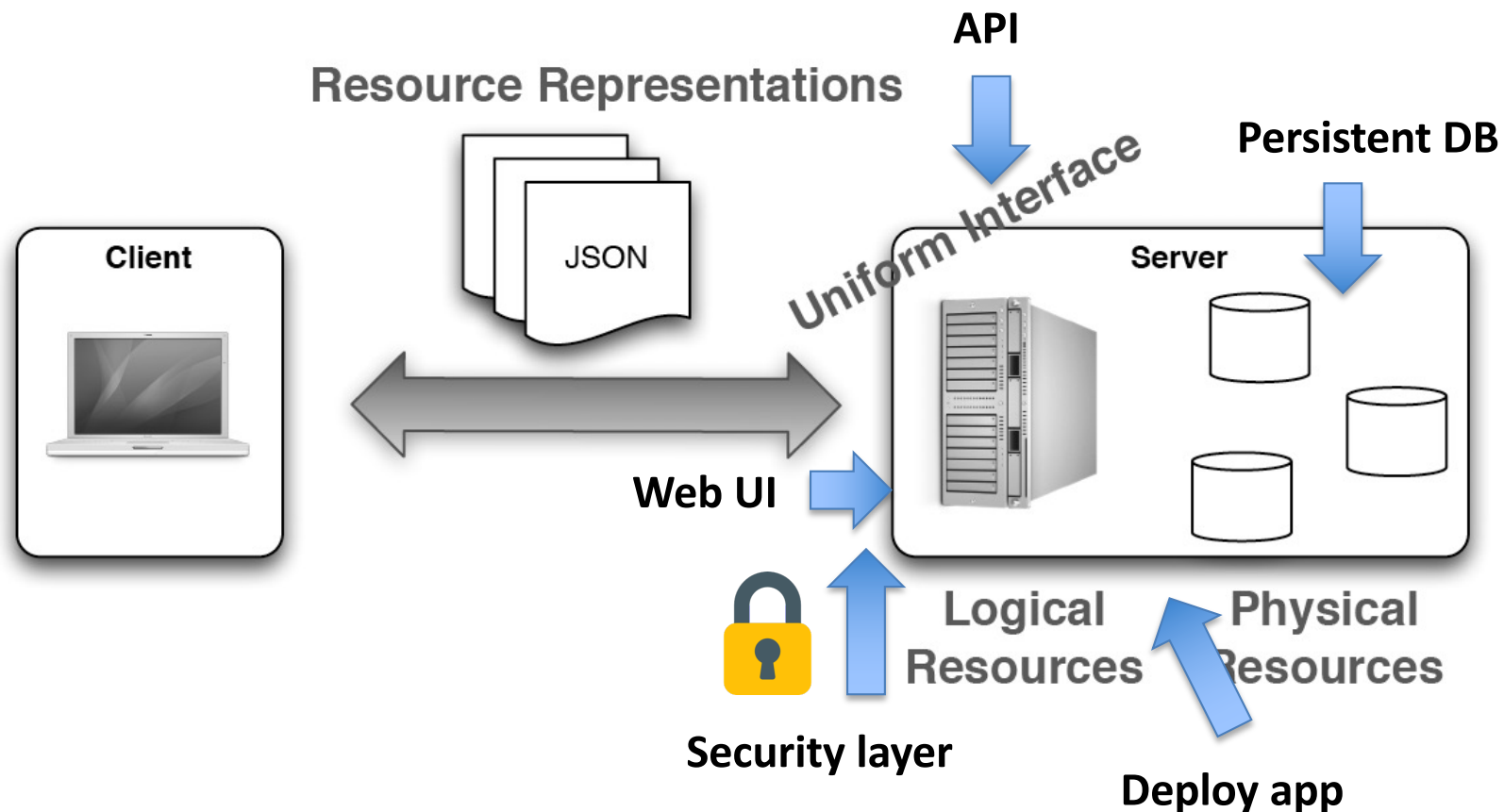
All parameters except [state](#) are required.

<https://developers.facebook.com/docs/instagram-basic-display-api/guides/getting-access-tokens-and-permissions>

Parameter	Sample Value	Description
client_id Required <i>Numeric string</i>	990602627938098	Your Instagram App ID displayed in App Dashboard > Products > Instagram > Basic Display .
redirect_uri Required <i>String</i>	https://socialsizzle.herokuapp.com/auth/	A URI where we will redirect users after they allow or deny permission request. Make sure this exactly matches one of the base URIs in your list of valid OAuth URIs . Keep in mind that the App Dashboard may have added a trailing slash to your URIs, so we recommend that you verify by checking the list.
response_type Required <i>String</i>	code	Set this value to <code>code</code> .
scope Required <i>Comma- or space-separated list</i>	user_profile,user_media	A comma-separated list, or URL-encoded space-separated list, of permissions to request from the app user. <code>user_profile</code> is required.
state <i>String</i>	1	An optional value indicating a server-specific state. For example, you can use this to protect against CSRF issues. We will include this parameter and value when redirecting the user back to you.

<https://developers.facebook.com/docs/instagram-basic-display-api/guides/getting-access-tokens-and-permissions>

General *reference architecture for our projects*



Software Versioning and collaboration

- **Software Versioning:** It's the process of assigning either unique version names or unique version numbers to unique states of computer software

unique version name/number <-> unique state

- **Collaborative software:** help people working together on a common task to attain their goals

Testing

“Testing is the process of executing a program with the intent of finding bugs”

– GLEN MYERS

“Testing can show the presence of bugs but never their absence”

– DIJKSTRA

Testing, test suite and coverage

- **Test suite:** collection of **test cases** to test a software program
- **Coverage** is a key metric in software testing

Test case

Exercise:

What is a test case?

Testing suite

Exercise:

Do you expect the testing suite is a small software package with respect to the lines of code (KSLOC) of your target application?

Note: KSLOC means thousands of "Source Lines Of Code" or, in other words, lines of code excluding blank lines and comments.

Testing example: SQLite

From SQLite Testing section Introduction:

*“As of version 3.33.0 (2020-08-14), the SQLite library consists of approximately **143.4 KSLOC** of C code. By comparison, the project has **640 times** as much test code and test scripts - **91911.0 KSLOC**.”*

- <https://sqlite.org/testing.html>

Note: KSLOC means thousands of "Source Lines Of Code" or, in other words, lines of code excluding blank lines and comments.

Deploy

Software deployment is about all of the activities that make a software system available for use.

- **Pre-production environment:** it resembles a production environment.
- **Production environment** is the real, target, operational environment
 - it needs an *operation manager*

DevOps

- *DevOps is a set of practices that **combines software development (Dev) and IT operations (Ops)***
- It aims to shorten the systems development life cycle and **provide continuous delivery** with high software quality
- We'll see some practical aspects during the hands-on classes

Feedback form

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